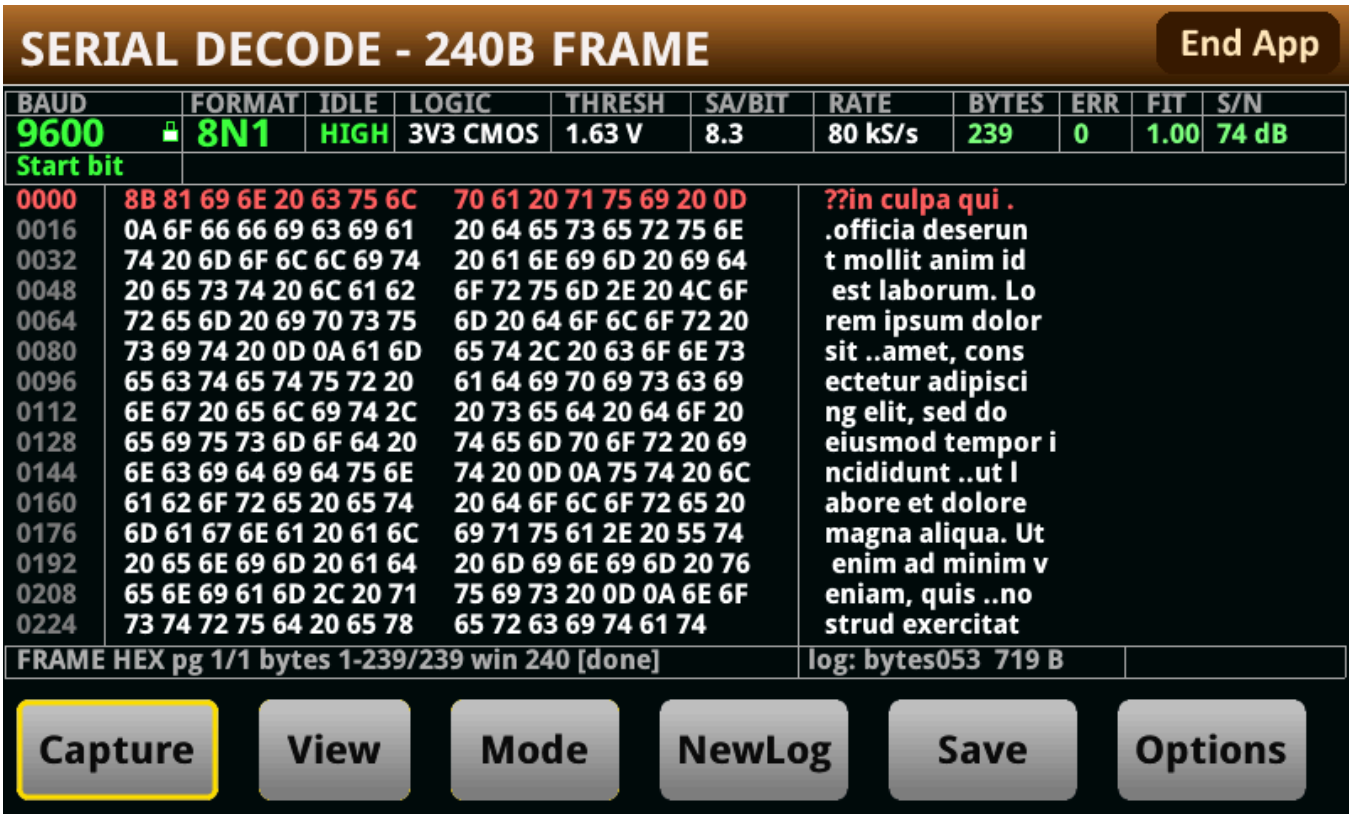
Serial Decode — Keithley DMM6500, DAQ6510, DMM7510, SMU2461

A UART decoder that runs **on** a Keithley bench instrument. It digitizes the line with the instrument's own digitizer, recovers the baud rate, frame format and idle polarity from the signal, and shows the bytes on the front panel as text or hex. No host, no logic analyser, one probe.

Ian Ameline · **version 1.25** · MIT licence (see LICENSE)



The instrument's own front panel, no host attached: 239 bytes of a 240-byte capture at 9600 baud 8N1, no framing errors, S/N 74 dB. The padlock on the BAUD cell means the rate is pinned here rather than re-detected each capture.

docs/MANUAL.md	using it: hooking up, the screen, the buttons, what it copes with, where it fails
docs/REFERENCE.md	every measured number, and the firmware limits behind them
docs/BENCH.md	the harness: the seeded sweep, the release gate stage by stage, reproducing a failure

Ships as `Serial_Decode.tspa`, installed from a USB key through the instrument's own **Manage Apps** screen. Written in TSP — Lua **5.0.2** embedded in the firmware, so the sources avoid `#`, `%`, `string.gmatch` and bitwise operators throughout.

What to know before trusting it

Tested on a DMM6500 and nothing else. Every measured number in these documents came off that one instrument. See Which instruments.

Flow control is verified electrically but never closed as a loop. One credit pulse per armed capture, measured on a scope, with nothing waiting for it on the other end. The width is not settable on this firmware, so a receiver needs an edge-triggered input rather than a polling loop. See **Triggering, in both directions** in REFERENCE.md.

Automatic rate detection has three known failures — 8 points in 860 on a full bench plan for the first two and about 3 in a million for the third — and all three are fixed by typing the rate in: a short pattern repeated over and over can measure at a small multiple of its true rate, a logic low near **–2 V** can misdetect, and above 115 200 baud a long run of one repeated byte can refuse. **None reaches an ordinary payload at a standard baud rate:** across 171 600 captures at standard rates, opened from 200 different points in the waveform, the fox, the 1 kB text payload, twelve random payloads and the walking-bit patterns are clean at every rate from 300 to 250 000 baud. All three are characterised under **Known failures of automatic rate detection** in the manual.

Nothing stops a long job early. A touch press cannot be delivered while a script runs, and the front-panel TRIGGER key does **not** deliver `trigger.EVENT_DISPLAY` during a panel-initiated run, so a recording runs to its stated size. That key does still *gate* an armed capture under `Options ▸ Trigger = Trigger key`: arm with Capture, then press TRIGGER and the acquisition fires on the key edge in firmware.

The TRIGGER key cannot be made to start a capture, and that was tried rather than assumed. A touch button works because the firmware executes a Lua string attached to a display *object*; the key is not an object, so nothing can be attached to it and all it can set is a latch. Reading a latch needs Lua running, and a running script is exactly what blocks the touch panel — so a poller would cost the whole glass, and only a touch can start Lua in the first place. Either the key works or the screen does, never both, and the screen is the one worth having.

One press is the whole interaction. A screenful is ~240 bytes; a recording holds 8 192 or 32 768 bytes to a file, decoded and filed without stepping; beyond that, flow control makes the window a chunk size and credits the device until it stops sending. Ceilings and arithmetic are in the manual.

It takes over the measurement function and gives it back. Capturing means digitizing, so every press of Capture selects **Digitize Voltage** on the 10 V range whatever the meter was measuring. The mode is read once at startup and restored when you press **End App**, so ending the app leaves the meter measuring what it was measuring when you launched it, on the range it was on. Two consequences, both deliberate: a function changed from the front panel *while the app is loaded* does not survive the exit — the next Capture reselects the digitizer, so the app keeps working — and the bench soak in `bench/` bypasses this path entirely, so a soak does leave the instrument digitizing.

Endurance, measured

Three unattended runs on real hardware underwrite everything below. All three are the instrument driving itself: it reads its plan from its own USB key, commands the generator over the LAN, decodes, judges and writes every cell back to the key. **No computer is attached to any of it.**

run	when	duration	cells judged	BAD	rate
the week	08-29 → 09-05	**182.2 h**	133 297	2 462	**1.85 %**
the 17-hour	09-07	17.24 h	11 739	143	**1.22 %**
the 8-hour	09-08	8.28 h	6 708	76	**1.13 %**

The week: seven and a half days unattended

182.2 hours, 136 247 captures, twelve segments, and not one instrument event. On this firmware an "event" is a modal box that nothing in the app can suppress, so it is the thing that ends an unattended run — and every segment's closing total reports **0 events instigated, 0 of them from the panel**, with **0 cells unrecorded in 0 gaps**. The USB key held the whole record; nothing was lost at a segment seam.

The generator was the only part that needed help: **270 waveform re-selections across the week, every one recovered** — 241 on the second attempt, 28 on the third, 1 on the fourth. That is a known SDG2122X behaviour, not an app fault: `C1:ARWV?` answers with the *previous* waveform's name on roughly 4 of every 39 real switches, and re-sending the selection fixes it while re-reading does not. It reproduced at exactly that rate in the 8-hour run below (10.3 %), so it is characterised rather than mysterious.

The week's 1.85 % is not comparable with the two runs below, and the drop is not a measured improvement. It ran an older build *and* a different matrix — laps of 1 326 and 1 333 cells with a different vector set, different amplitudes and different waits. Two numbers from different plans are two different questions.

The two current-matrix runs

Both are the 1 677-point matrix on the current build, each on a single power cycle: **7 laps in 17.24 h** and **4 laps in 8.28 h**. Pooled, they give **219 BAD in 18 447 cells — 1.19 %** — and they agree with each other **within 0.6 σ**, which is what lets them be pooled at all.

Split by rate, the 8-hour run reads **1.28 %** at or below 115 200 Bd and **0.75 %** above it; the 17-hour run **1.34 %** and **0.89 %**. The high-rate band is the better half in both, which is the opposite of the naive expectation and falls out of there being more samples per bit to disagree about at the low end.

The 8-hour run is the cleanest record on file: **0 no-decode, 0 inconclusive, 0 generator failures, 0 events instigated, 0 cells unrecorded in 0 gaps**, ending `4 iteration(s) complete` under its own control rather than being stopped.

What a low single-digit failure rate means here

It is the expected shape, because the vectors are built to break things. Impulse spikes stacking to 9.3 V on a 3.3 V line; noise and a second signal riding on the logic, summed in from the generator's CH2 at a sweepable level; baud drifting 6 % and 10 % mid-waveform; 2, 10 and 20 % jitter; an inverted line; LIN break fields no UART frame can legally contain; 256- and 512-byte runs of a single value with almost no transitions; payloads longer than the capture. **Every one of them is swept across all 43 rates**, including rates two decades away from the one it was built for.

Composition is what stops that being an excuse. Only the seven vectors built to break something are allowed to fail — `v47`, `v48a/b`, `j20`, `v61/62/63`, class `loud` in `tools/soakplan.py` — and they carry **39 % of the current matrix's failures; the other 61 % are `exact` vectors, where a failure is a defect** and is counted as one (86 and 132 of 218 pooled; the week's split was 32/68). **A confidently wrong byte fails every class**, `loud` included: declining to answer is a pass, being wrong while claiming to be right never is.

Where the app *declines*, it is concentrated and explicable. In the whole 8-hour run only three vectors refuse a cell at all — `v48b` 168, `v48a` 158, `v47` 32 — and the drift vectors' reason is `no clear logic levels`, which is the correct answer to a band that has moved onto its own threshold.

Confidently wrong bytes, across all three runs: none. Every failure is the app reporting a rate its own framing contradicts, or declining to answer. Separately, over **1 066 400 offline decodes: 95.9 % of failures are the rate being wrong and 0.026 % get a byte wrong with the rate right** — see **Rate detection and decode, counted separately** in REFERENCE.md.

Nothing leaked, and lap time did not rise

Heap after collection, read at every lap seam: the 17-hour run gave **2678, 2678, 2679, 2678, 2679, 2678, 2678 kB — flat to 1 kB over 10 062 cells**, and the 8-hour run **2687, 2687, 2687 kB — flat to 0 kB over 5 031 cells**. Lap time did not rise in either, which catches a leak that no failure count can see.

One counter did drift, in the 17-hour run and in a quantity nobody was watching: `v47`'s refusals rose 7, 7, 7, 9, 9, 11, 14 across its seven laps — monotone, 5.1 σ. **It did not reproduce**: the 8-hour run gave 9, 9, 6, 8, flat and non-monotone. Four laps is short against a claimed +1.11 per lap, so this neither confirms nor kills it; it does mean it is not reliably present. It is recorded rather than explained.

One bench hazard worth naming, because it is not the app

`acq: line is idle (no transitions)` is accumulated state in the instrument, and it can cost an unattended run. A soak started shortly after the 13-minute smoke gate hit it hard: 33 idle cells in its first 49, with the generator reading back the correct waveform throughout and the record showing `SDG failed 0`. It presents as a generator fault and is not one — power-cycling the generator changes nothing, power-cycling the **DMM** clears it, and 8.3 hours of running afterwards did not bring it back.

So: **power cycle the DMM between the smoke gate and a soak.** The cheap tell that you have it is cell *time* — an idle cell fails in about 1 s against a healthy 5.4–6.4 s, so a poisoned run visibly races. It costs a bench run, not correctness: an idle cell is a refusal, not a wrong answer. Details under *Instrument hazards* in BENCH.md.

How the offline twin compares, and why it is no longer the safe side

The offline twin is checked against these runs rather than trusted — **and the comparison only means anything at a matched stimulus.** Its default soak deliberately drives half its cells outside the generator's envelope, which a current bench lap never does, so the two have to be set side by side on the same plan. Matched, over **444 405 cells in 265 laps** it fails on **1.135 %** against the bench's **1.187 %** pooled over 11 laps — **0.96×, which is 0.67 σ from parity**. Earlier readings of "over-predicting by 1.48×" compared a stressed harness against an unstressed bench and are withdrawn.

That parity is partly two errors cancelling, and the per-vector table in BENCH.md is what says so: the twin runs four vectors about 1.3× hot and invents ~1.5 failures a lap on random payloads the bench never fails, while scoring zero on the two drift vectors — where it is in fact the *harsher* side, declining 100 % of those cells at the acquisition gate against the bench's 92–98 %. **So the twin is no longer safely pessimistic**, which is the property this project wants of it, and that is worth knowing before leaning on it as a gate by itself.

Which instruments

The app needs a digitizer reachable as `dmm.digitize` or `smu.digitize`, the **touchscreen app API** (`display.create` and friends), and **Lua 5.0.2**. Every digitizer below runs at **1 MS/s**, and the **panel is pixel-identical across the whole TSP range**, so the app's screens carry over as they are.

DMM6500	**Tested**, firmware 1.7.17a — a 7.6-day run of 136 247 captures with no computer attached and zero instrument events, then 17.24-hour and 8.28-hour runs on the current matrix, and the namespace resolver verified here. 16-bit digitizer, "*maximum resolution 16 bits*", specifications, April 2018
DAQ6510	Should run unmodified, on the strongest grounds of any untested model: it **shares the UI board and the acquisition board** with the DMM6500, only the channel-board plugin differing. Untested
DMM7510	Should run unmodified: 18-bit digitizer, better acquisition boards. Untested
SMU2461	**Will install and try; may or may not work.** Dual 18-bit digitizers, reached as `smu.digitize` by a namespace the app resolves at load. That mechanism is verified on the DMM6500; no SMU has ever run it. Three unknowns below
2470 SMU	**No.** "*Digitized measurements are not a feature on the 2470*" — its reference manual, rev D, October 2024
2450 / 2460 SMU	Almost certainly not, by absence rather than denial: neither claims a digitizer, and the 2450 reference (rev D, May 2015) documents `smu.measure.*` with no digitize function

Porting is an acquisition question only. The panel is identical, so the display layer carries over untouched, and the app resolves `dmm.digitize` or `smu.digitize` once at load with no test on any capture path. Every rate figure in REFERENCE descends from the sample rate and the reading buffer's throughput, so the timing figures transfer; resolution is the one axis the app is indifferent to, thresholding each sample into a one or a zero. What to re-measure is what goes *through the acquisition board* — the tolerance envelope, the level and offset limits, the –2 V band. The DAQ6510 shares that board, so a deviation there is a real finding; the DMM7510's differs and those figures may move either way.

On an SMU2461 the app will install and try, and three things could stop it. *What the reading buffer holds* — a 2461 digitizes voltage and current **simultaneously** and the decoder trusts consecutive indices, so a buffer carrying current or source values alongside voltage reads as a waveform and surfaces as garbage bytes rather than an error. *The range* — the app pins **10 V**, chosen for the widest digitize bandwidth, which a 2461 may not offer. *The source output* — the app sets it off before every capture and **reads it back, refusing the capture if it cannot confirm it**; those constant names are unverified, and that refusal is the failure it is designed for.

The `.tspa` header decides where it installs, independently of whether the code would run: `$Product: DMM6500, DAQ6510, DMM7510, SMU2461`. Neither `DMM7510` nor `SMU2461` appears in the value set Keithley's app-header spec publishes, though Keithley ship TSP apps of their own declaring DMM7510 — an install failure is the symptom if a firmware validates the field strictly, and this field is the first thing to revert.

If you run this on anything but a DMM6500, please say so — open an issue. Every row above except the first is inference from API surfaces and vendor documents, and one report from real hardware outweighs all of it. Useful to include: `localnode.model` and `localnode.version`, whether **Manage Apps** offered the app at all, whether both screens built, and whether a capture decoded.

How to report bugs

docs/BETA.md (PDF) is the note to read first, and the one to hand to anybody trying this on their own bench. It lists the four behaviours that **look** like defects and are not — chiefly that a signal whose band straddles ground is decoded inverted, deliberately — so a tester does not spend their evening rediscovering them.

Then open an issue with, as far as you can: a **screen grab** of the whole 800x480 frame (`tools/screenshot.py` takes one over LXI with nothing loaded), the **text that should have been decoded**, the **bit rate** you set, and the **framing** you sent. The expected payload is the one piece that cannot be reconstructed from anything else — it turns "the decode looks wrong" into a difference that can be computed.

A wrong number that looks plausible is worth more than a crash.

Before releasing it to anyone

Three gates, each about ten times the cost of the one before it. Run them in order.

```
python3 tools/release_sweep.py --offline   #  ~1 min, no instruments
python3 tools/bench_smoke.py               #  13 min, both instruments
                                           #  then POWER CYCLE THE DMM -- see below
python3 tools/soak.py --hours 17 --suites formats,plan --skip-vectors v95,v96,v97
python3 tools/release_sweep.py             #  the whole thing, instruments included
```

Name all three skipped vectors, or the run is not the run you think it is. `soak.py`'s `--skip-vectors` defaults to empty and the skip is applied *before* the shuffle, so it sets the lap size and keys every amplitude, offset and wait: `v95,v96,v97` gives a **1 677**-cell lap, and naming only `v95,v96` gives **1 720** and lets `v97` back in, where it reads as a generator wedge — that is where an earlier run's 129 "generator failures" came from. `run_bench.py` is the opposite: it already defaults to all three, so do not pass the flag to it at all.

Power cycle the DMM between the smoke gate and the soak. A soak started shortly after a smoke can hit the accumulated-state failure described under *Endurance* above; it costs the run, not correctness.

The smoke gate covers the whole rate range in 13 minutes. Four waveforms — the fox and a random payload, each in 8N1 and 7E1 — paired crosswise, so one pair takes the 22 standard baud rates and the other the 21 drawn ones, and each rate family faces both formats and both content classes. Then all 45 button presses, seven logic swings from 5 V down to 0.25 V, and eight wrong-locked-rate cases. Measured: 86 cells in 9.7 min, 45 presses in 1.9, 7 levels in 0.7, 8 rate cases in 0.9.

No version is tagged without at least 8 hours of soak. A soak reports a **failure rate per test point**, where a sweep reports one pass or one failure. A lap is 1 677 cells and about 2.1–2.5 hours, so 8 hours is the least that separates "fails every lap" from "failed once"; 17 hours gave seven laps and 8.3 gave four.

Code changes do not get pushed without passing the smoke gate. On a pass, `bench_smoke.py` writes a receipt holding a hash of `tsp/` and `tools/`; a `pre-push` hook recomputes it and refuses if either tree has moved since. Documentation-only pushes are unaffected — neither tree is hashed — and when the bench is unavailable, `SMOKE_OVERRIDE="reason" git push` proceeds and prints the reason.

```
ln -sf ../../tools/hooks/pre-push .git/hooks/pre-push
```

Every lap draws its waveform order, its non-standard rates and the wait before each capture from the lap number, so `--iteration N` rebuilds any lap exactly without running the ones before it, and a failure replays offline in seconds.

docs/BENCH.md has the rest: every stage of the release gate and what it checks, the auditable record a full sweep leaves behind, what state the instruments have to be in before it will start, how to replay a failing cell offline, and the hazards worth knowing.

Layout

`tsp/`	the app. `serial_core` acquisition, `uart_decode` framing, `chunk_decode` resumable decode, `serial_ui` panel, `serial_app` orchestration
`bench/`	the soak that runs **on** the instrument, loaded beside the app for a multi-day run and never shipped inside it. bench/README.md is the runbook
`tools/`	harnesses. `release_sweep.py` is the entry point; the rest are the authorities it calls
`docs/`	the manual, instrument references, panel mockups, and BETA.md for testers

`tsp/midi_decode.tsp` and `tsp/lin_decode.tsp` are complete and tested but **not shipped** in version 1 — the LIN checksum has never been checked against a real frame. Re-adding either is one line in `tools/package_tspa.py`; the app discovers what it has at runtime.

Bench

Two instruments over LAN: the DMM6500 under test and an SDG2000X-series generator producing the stimulus. **A scope is optional** — an SDS1000X-E is used here to confirm the stimulus is what it claims to be, and nothing in the gate needs it; it only ever reads. `tools/instruments.py` holds the addresses and the hazards: repeated large waveform uploads wedge the generator's LAN service, and calibration state is off limits on both.

Neither line's top model is needed. The bench asks the generator for 25 MSa/s of TrueArb, 20 Vpp and ~41 stored waveforms, and the scope, if used, for UART decode, two buses and 1 GSa/s; in both lines the model number is the *analog bandwidth*, and the fastest edge here is a 250 kBd bit. An **SDG2042X (40 MHz) and an SDS1104X-E (100 MHz) are sufficient**; the units here are an SDG2122X and an SDS1204X-E. `SDG_MAX_SRATE` is capped at 40 MSa/s so a plan cannot quietly outgrow the cheapest generator; see docs/BENCH.md for what to check on a unit that is not the one on this bench.

The generator must be 2000X-series or better, and an SDG1000X will NOT do — worth stating because the scope here is a 1000X-E and the numbers invite the wrong substitution. TrueArb is selected by the `SRATE` command, and Siglent's programming guide lists its availability per family:

	SDG800	SDG1000	SDG2000X	SDG5000	SDG1000X	SDG6000X/X-E	SDG7000A
`SRATE`	no	no	**yes**	no	**no**	yes	yes

An SDG1000X stores and plays arbitrary waveforms, but DDS-only — and DDS **resamples** the stored points, which is precisely the fall-back `SDG.assert_truearb()` checks for on every load. It would not merely be less accurate; the vectors would stop being valid stimulus, because sub-sample edge timing is the quantity this project measures. Note also that `INTER`, the interpolation-method control, is `no` even on the SDG2000X — reachable only on the 6000X/7000A — which is why a 2000X's TrueArb output is a plain held staircase with nothing to switch off.